

## Practice problems #9

1. A dietician is planning a meal that supplies certain quantities of vitamin C, calcium, and magnesium. Three foods will be used, their quantities measured in appropriate units. The nutrients supplied by these foods and the dietary requirements are given here.

| Nutrient  | Food 1 | Food 2 | Food 3 | Required |
|-----------|--------|--------|--------|----------|
| Vitamin C | 10 mg  | 20 mg  | 20 mg  | 100      |
| Calcium   | 50 mg  | 40 mg  | 10 mg  | 300      |
| Magnesium | 30 mg  | 10 mg  | 40 mg  | 200      |

Write a vector equation for this problem. State what the variables represent, and then solve the equation.

2. Given  $A = \begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & 3 \\ 9 & -3 & 3 \end{bmatrix}$ , note that  $a_1 + 2a_2 = a_3$ . Here,  $a_i$  denotes the  $i$ -th column of the matrix  $A$ .

*Without any calculation*, answer the questions:

- (a) How many pivot positions of  $A$  do you expect? Why?
- (b) What is a geometric description of the span of the columns of  $A$ ?
- (c) Do the columns of  $A$  span  $\mathbf{R}^3$ ? Why or why not?

Answers are on page 2.

#1:  $x_1 \begin{bmatrix} 10 \\ 50 \\ 30 \end{bmatrix} + x_2 \begin{bmatrix} 20 \\ 40 \\ 10 \end{bmatrix} + x_3 \begin{bmatrix} 20 \\ 10 \\ 40 \end{bmatrix} = \begin{bmatrix} 100 \\ 300 \\ 200 \end{bmatrix}$ , where  $x_1, x_2$  and  $x_3$  are the number of units of foods 1, 2, and 3, respectively, to be used in the meal.  
The solution is  $x_1 = 150/33, x_2 = 50/33$ , and  $x_3 = 40/33$ .

#2(a): 2. Why?

#2(b): A plane in where? Plane contains what?

#2(c): No. Why not?